Course Packaging Analysis

Crispin Jordan, Nicola Romano, Franziska McManus

2026-08-31

```
knitr::opts_chunk$set(echo = TRUE)

library(here) #here() loads data
```

```
## here() starts at /Users/crispinjordan/Documents/TeachingFellowshipBioMed/PTAS_application2021/PTAS_An
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(ggtext)
library(janitor) # for clean_names()
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```
library(gridExtra) #grid.arrange() displays multiple plots
```

```
##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
library(scales) #label_wrap() displays axis labels over multiple lines
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
```

```
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(brms)
```

```
## Loading required package: Rcpp
## Loading 'brms' package (version 2.23.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').
##
## Attaching package: 'brms'
##
## The following object is masked from 'package:stats':
##
##     ar
```

```
library(broom.mixed) #Used in prior sensitivity analysis
library(ggrepel)  #Used in prior sensitivity analysis

set.seed=1234
```

Load **survey_data**

- survey responses from qualtrics
- includes: 4c) How does the degree package its instruction in Data Analysis and/or quantitative Study Design?

```
#load survey data
survey_data <- read_rds(here("data/clean_data", "survey_data.rds"))

#look at data
survey_data %>%
  select(university_id, discipline, course_packaging) %>%
  glimpse()
```

```
## Rows: 103
## Columns: 3
## $ university_id    <fct> u90, u90, u67, u67, u66, u67, u97, u86, u93, u12, u87~
## $ discipline       <fct> Bio, Psych, Bio, Med, Bio, Psych, Biomed, Biomed, Med~
## $ course_packaging <ord> Both, Both, Both, StandAlone, Both, Both, Both, Both,~
```

```
#structure: ordered factor
str(survey_data$course_packaging)
```

```
##  Ord.factor w/ 3 levels "Integrated"<"Both"<..: 2 2 2 3 2 2 2 2 2 2 ...
```

```
#table: course packaging by discipline
table(survey_data$discipline, survey_data$course_packaging)
```

```
##
##          Integrated Both StandAlone
##   Bio             8   17          6
##   Biomed          7    7          5
##   Med             8    3          4
```

```
##   Psych           2   16         20
```

Multinomial analysis

We now fit a multinomial model, using a prior recommended by Gelman et al. (2008) (DOI: 10.1214/08-AOAS191).

Below, we (1) identify model components that require a prior, (2) specify the prior, (3) specify contrasts (as recommended by Gelman el al. (2008)), (4) perform a prior predictive check.

```
get_prior(course_packaging ~ discipline, family = categorical(link = "logit"), data = survey_data)
```

```
##                 prior     class             coef group resp         dpar nlpar
##                (flat)         b                                    muBoth
##                (flat)         b disciplineBiomed                   muBoth
##                (flat)         b    disciplineMed                   muBoth
##                (flat)         b  disciplinePsych                   muBoth
##  student_t(3, 0, 2.5) Intercept                                    muBoth
##                (flat)         b                               muStandAlone
##                (flat)         b disciplineBiomed              muStandAlone
##                (flat)         b    disciplineMed              muStandAlone
##                (flat)         b  disciplinePsych              muStandAlone
##  student_t(3, 0, 2.5) Intercept                                 muStandAlone
##  lb ub tag       source
##                 default
##            (vectorized)
##            (vectorized)
##            (vectorized)
##                 default
##                 default
##            (vectorized)
##            (vectorized)
##            (vectorized)
##                 default
```

```
#Our prior, following Gelman et al. (2008):
priors.package <- c(
  prior(cauchy(0, 10), class = "Intercept", dpar = "muBoth"),
  prior(cauchy(0, 10), class = "Intercept", dpar = "muStandAlone"),
  prior(cauchy(0, 2.5), class = "b", dpar = "muBoth"),
  prior(cauchy(0, 2.5), class = "b", dpar = "muStandAlone")
)


contrasts(survey_data$discipline) =contr.sum(4)
contrasts(survey_data$discipline)
```

```
##        [,1] [,2] [,3]
## Bio       1    0    0
## Biomed    0    1    0
## Med       0    0    1
## Psych    -1   -1   -1
```

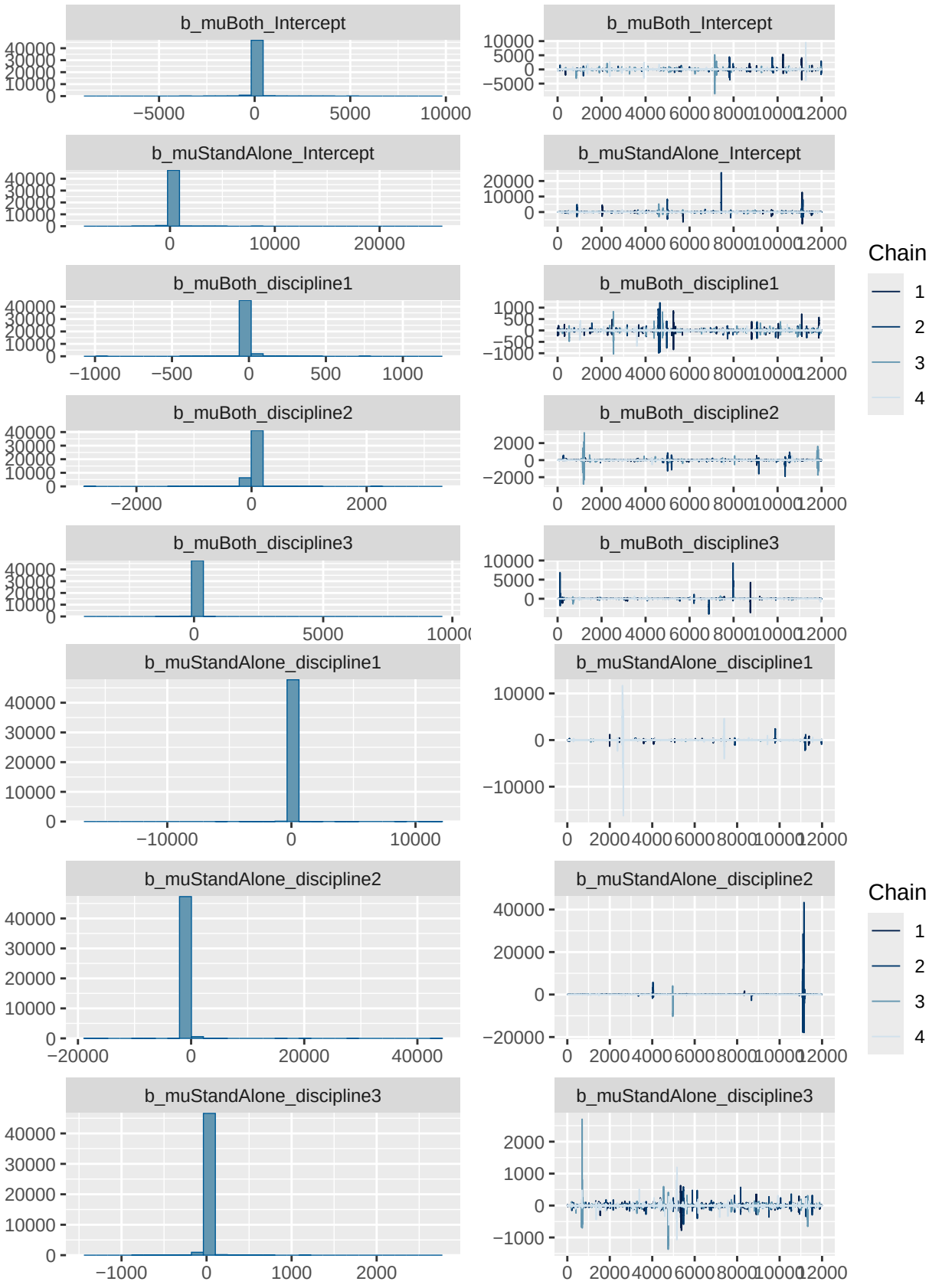
```
#Prior predictive check:
brm.formula.packaging <- brmsformula(course_packaging ~ discipline,
                     family = categorical(link = "logit"))
```

```
brm.mn.package.priorpc <- brm(brm.formula.packaging,
                              prior = priors.package, sample_prior = "only",
                              iter = 24000,
                              control = list(adapt_delta = 0.999, stepsize = 0.001,
                                    max_treedepth = 20),
                              data = survey_data,
                              file = here("models.gelman", "brm.mn.package.priorpc"))

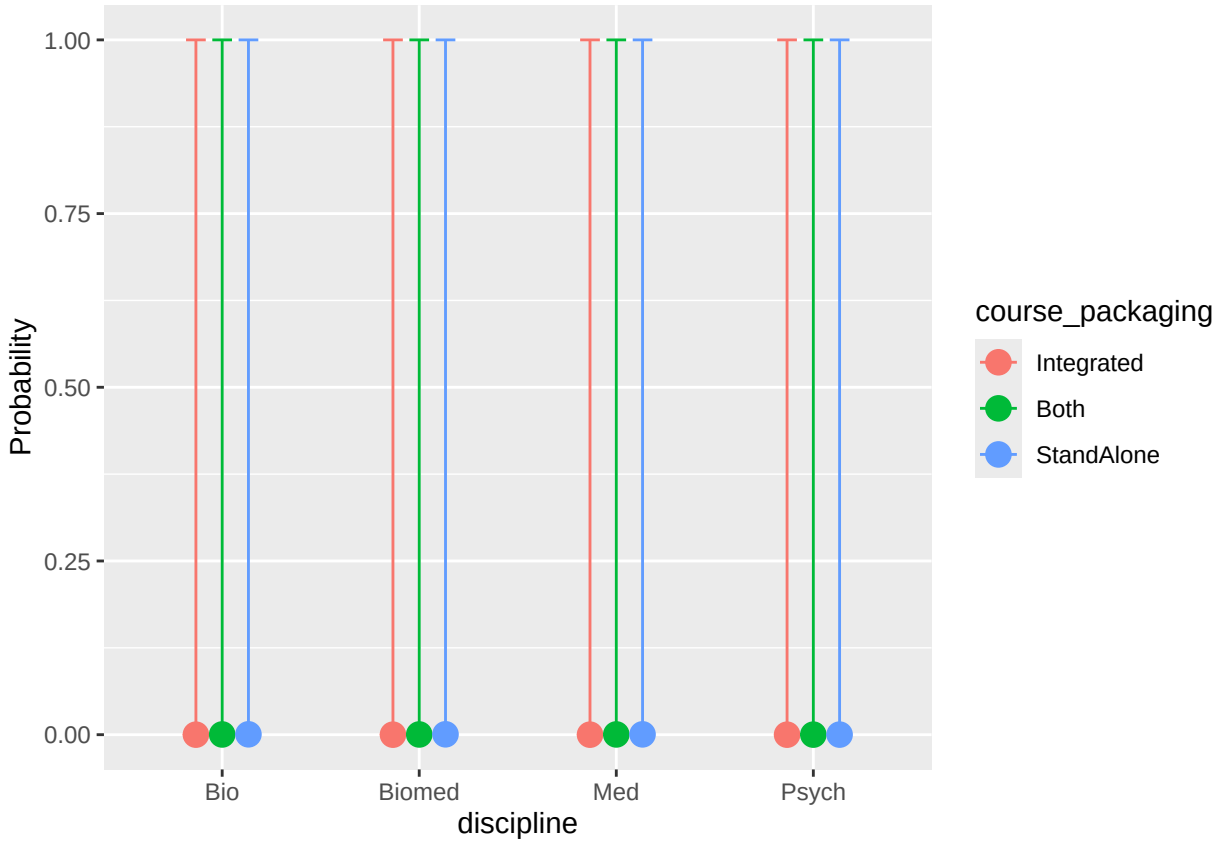
summary(brm.mn.package.priorpc)
```

```
##  Family: categorical
##   Links: muBoth = logit; muStandAlone = logit
## Formula: course_packaging ~ discipline
##    Data: survey_data (Number of observations: 103)
##   Draws: 4 chains, each with iter = 24000; warmup = 12000; thin = 1;
##          total post-warmup draws = 48000
##
## Regression Coefficients:
##                           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## muBoth_Intercept              1.66    237.67  -144.28   132.10 1.00    19187
## muStandAlone_Intercept        9.46    430.67  -143.01   158.47 1.00    13520
## muBoth_discipline1           -0.06     44.09   -28.20    28.68 1.00    25638
## muBoth_discipline2           -1.52    103.07   -37.82    35.77 1.00    11347
## muBoth_discipline3            0.40    117.33   -32.66    29.58 1.00    24000
## muStandAlone_discipline1      1.86    314.54   -37.05    35.90 1.00    15648
## muStandAlone_discipline2     13.95    906.67   -33.83    32.83 1.00    15662
## muStandAlone_discipline3     -0.46     48.04   -35.55    29.38 1.00    26839
##                           Tail_ESS
## muBoth_Intercept              5382
## muStandAlone_Intercept        3945
## muBoth_discipline1            7006
## muBoth_discipline2            3529
## muBoth_discipline3            6507
## muStandAlone_discipline1      3985
## muStandAlone_discipline2      4038
## muStandAlone_discipline3      5895
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(brm.mn.package.priorpc)
```


b_muBoth_Intercept
b_muStandAlone_Intercept
b_muBoth_discipline1
b_muBoth_discipline2
b_muBoth_discipline3
b_muStandAlone_discipline1
b_muStandAlone_discipline2
b_muStandAlone_discipline3
Chain
1
2
3
4

```
#Prior predictive check outout:
conditional_effects(brm.mn.package.priorpc, categorical = TRUE)
```



Output from `summary(brm.mn.package.priorpc)` and `plot(brm.mn.package.priorpc)` both indicate that estimates are centered on zero (the prior has little influence on estimates).

NOTE: exploratory analyses (not shown) indicate that stronger priors yield estimates that approach 33% for each discipline (with narrower credible intervals) in prior predictive checks (`conditional_effects(brm.mn.package.priorpc, categorical = TRUE)`).

Now we run the multinomial model with our specified prior, and check for appropriate model behaviour:

```
#Run the model with the our prior:
brm.mn.package <- brm(course_packaging ~ discipline,
                      family = categorical(link = "logit"),
                      prior = priors.package,
                      iter = 6000,
                      control = list(adapt_delta = 0.999, stepsize = 0.001,
                                     max_treedepth = 20),
                      data = survey_data,
                      file = here("models.gelman", "brm.mn.package"))

#Examine prior that we specified:
prior_summary(brm.mn.package)
```

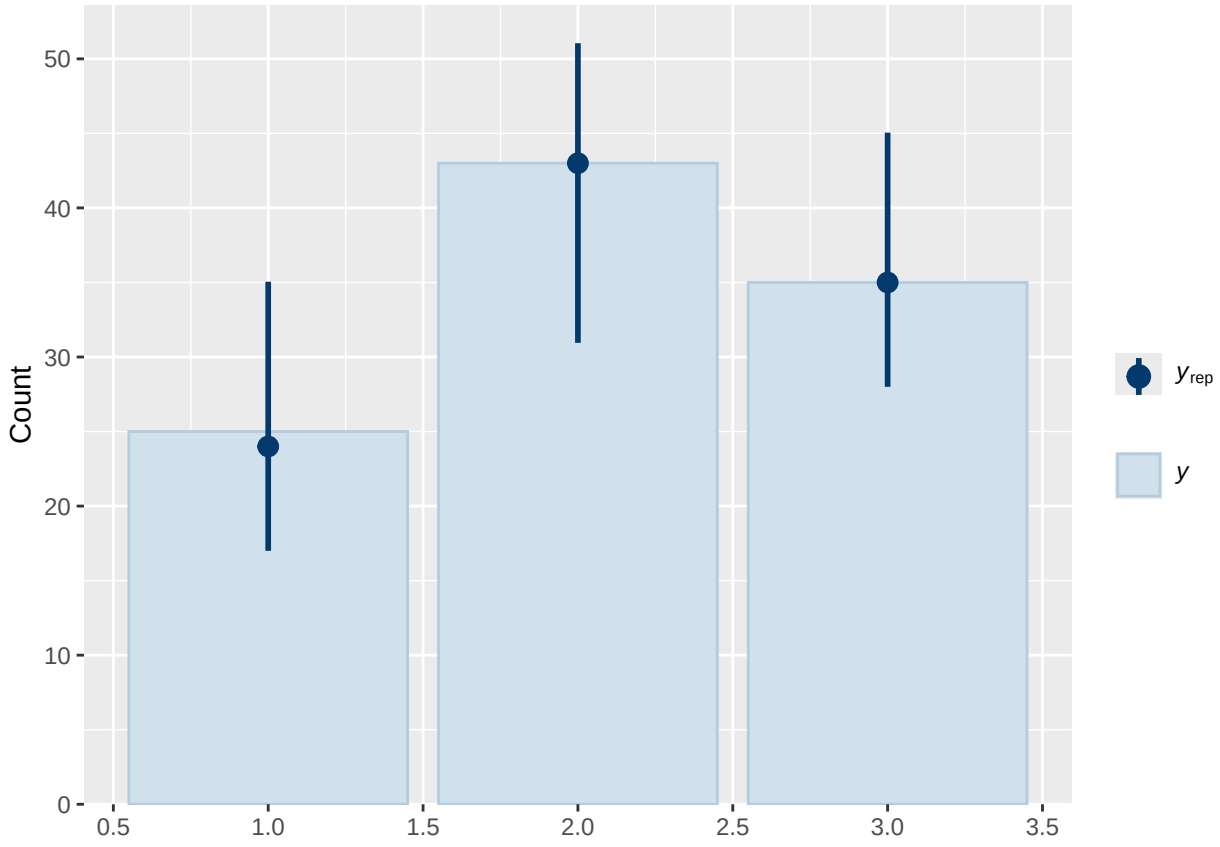
```
##           prior     class        coef group resp         dpar nlpar lb ub tag
##  cauchy(0, 2.5)         b                              muBoth
##  cauchy(0, 2.5)         b disciplineBio                muBoth
##  cauchy(0, 2.5)         b disciplinePsych              muBoth
```

```
##  cauchy(0, 2.5)         b discipline3                muBoth
##  cauchy(0, 2.5)         b                         muStandAlone
##  cauchy(0, 2.5)         b discipline1             muStandAlone
##  cauchy(0, 2.5)         b discipline2             muStandAlone
##  cauchy(0, 2.5)         b discipline3             muStandAlone
##   cauchy(0, 10) Intercept                              muBoth
##   cauchy(0, 10) Intercept                           muStandAlone
##        source
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##          user
```

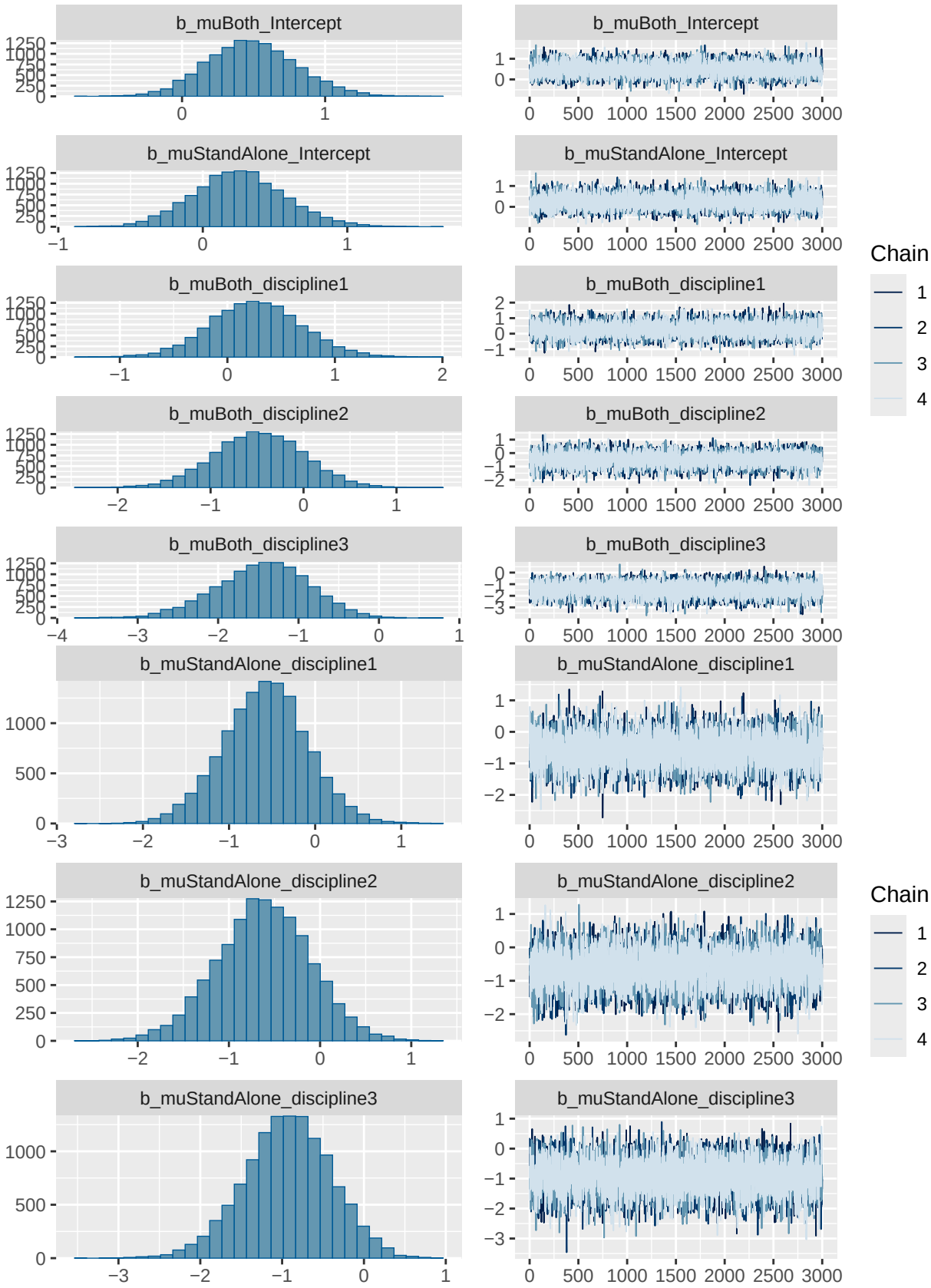
```
#Check convergence and coverage:
#We want Rhat to be less than 1.01
#We want _ESS to be 100 times the number of chains we run
summary(brm.mn.package)
```

```
##  Family: categorical
##   Links: muBoth = logit; muStandAlone = logit
## Formula: course_packaging ~ discipline
##    Data: survey_data (Number of observations: 103)
##   Draws: 4 chains, each with iter = 6000; warmup = 3000; thin = 1;
##          total post-warmup draws = 12000
##
## Regression Coefficients:
##                           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## muBoth_Intercept              0.49      0.31    -0.10     1.13 1.00     6347
## muStandAlone_Intercept        0.25      0.32    -0.35     0.91 1.00     7547
## muBoth_discipline1            0.27      0.42    -0.56     1.12 1.00     7447
## muBoth_discipline2           -0.48      0.48    -1.44     0.45 1.00     8384
## muBoth_discipline3           -1.45      0.57    -2.64    -0.38 1.00     8004
## muStandAlone_discipline1     -0.58      0.49    -1.56     0.37 1.00     7652
## muStandAlone_discipline2     -0.62      0.51    -1.65     0.38 1.00     7950
## muStandAlone_discipline3     -0.93      0.54    -2.02     0.09 1.00     6985
##                           Tail_ESS
## muBoth_Intercept              6629
## muStandAlone_Intercept        7545
## muBoth_discipline1            7521
## muBoth_discipline2            7903
## muBoth_discipline3            7720
## muStandAlone_discipline1      7417
## muStandAlone_discipline2      7914
## muStandAlone_discipline3      7407
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
#Posterior predictive check:
pp_check(brm.mn.package, type = "bars", ndraws = 100)
```




```
#Check traces
plot(brm.mn.package)
```



The summary (`summary(brm.mn.package)`) indicates that the model converged well (based on Rhat) and

has high coverage for all estimates.

The posterior predictive check output (`pp_check(brm.mn.package, type = "bars", ndraws = 100)`) indicates sufficient model fit.

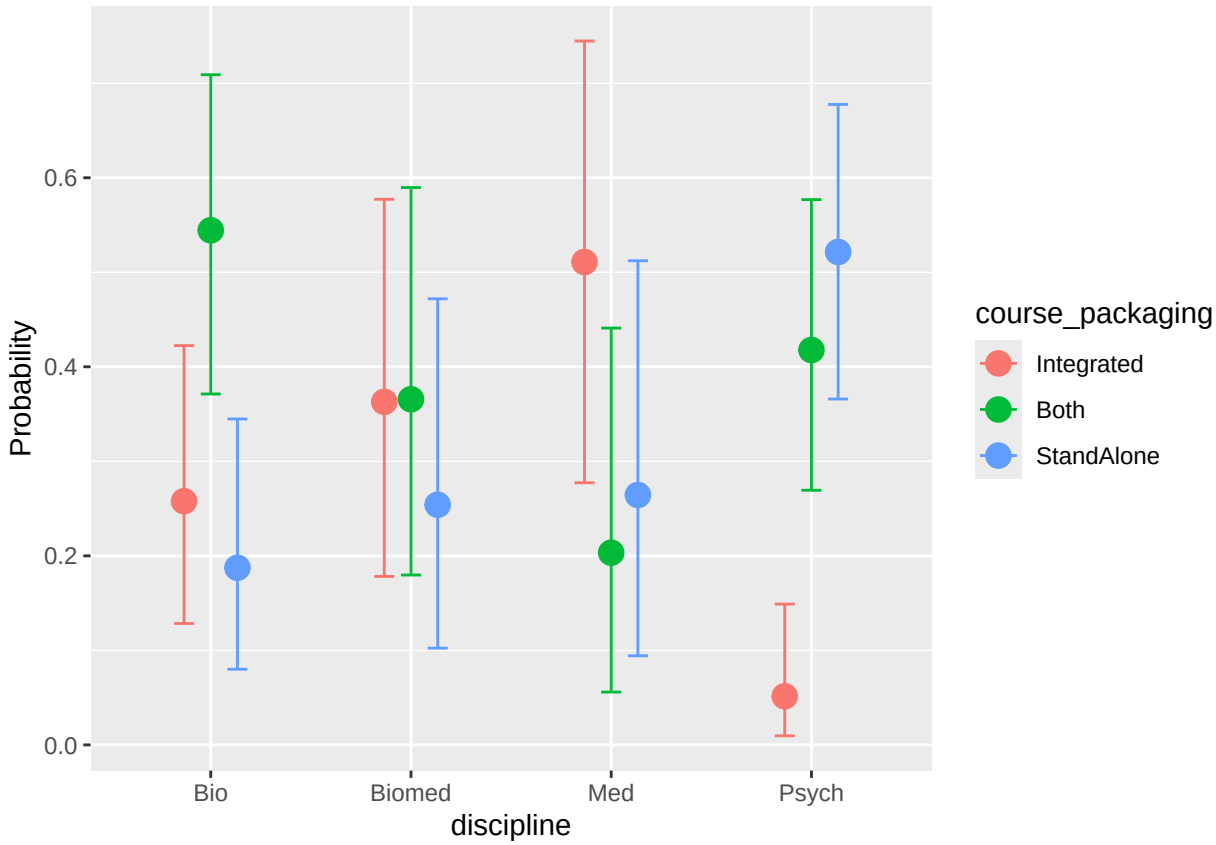
The traces and distributions of estimates on the latent scale (`plot(brm.mn.package)`) also indicate good model behaviour.

Now we obtain the model estimates:

```
eff_brm.mn.package <- conditional_effects(brm.mn.package, categorical = TRUE)
eff_brm.mn.package[[1]]
```

```
##    discipline course_packaging cond__     cats__ effect1__  effect2__
## 1         Bio       Integrated      1 Integrated       Bio Integrated
## 2      Biomed       Integrated      1 Integrated    Biomed Integrated
## 3         Med       Integrated      1 Integrated       Med Integrated
## 4       Psych       Integrated      1 Integrated     Psych Integrated
## 5         Bio       Integrated      1       Both       Bio       Both
## 6      Biomed       Integrated      1       Both    Biomed       Both
## 7         Med       Integrated      1       Both       Med       Both
## 8       Psych       Integrated      1       Both     Psych       Both
## 9         Bio       Integrated      1 StandAlone       Bio StandAlone
## 10     Biomed       Integrated      1 StandAlone    Biomed StandAlone
## 11        Med       Integrated      1 StandAlone       Med StandAlone
## 12      Psych       Integrated      1 StandAlone     Psych StandAlone
##    estimate__       se__    lower__   upper__
## 1  0.25783060 0.07537802 0.12845570 0.4224629
## 2  0.36295933 0.10445764 0.17822789 0.5772315
## 3  0.51088683 0.12597477 0.27733554 0.7446878
## 4  0.05147568 0.03362350 0.00960613 0.1490096
## 5  0.54444177 0.08764682 0.37126957 0.7090329
## 6  0.36568004 0.10883785 0.17971981 0.5896562
## 7  0.20335370 0.10237097 0.05591315 0.4410128
## 8  0.41777798 0.08089512 0.26941088 0.5769172
## 9  0.18740504 0.06659182 0.08006153 0.3448246
## 10 0.25404688 0.09812519 0.10235398 0.4719898
## 11 0.26437928 0.11140064 0.09430945 0.5122189
## 12 0.52150200 0.08118040 0.36592143 0.6775902
```

```
print(eff_brm.mn.package)
```



Plot

```
plot_colour <- "skyblue4"

plot_theme <- theme_bw() +
  theme(
    plot.title = element_text(size = 18),
    axis.title.x = element_text(size = 18),
    axis.text.x = element_text(size= 16),
    axis.title.y = element_text(size= 24),
    axis.text.y = element_markdown(size = 16, hjust = 1),
    axis.ticks.y = element_blank(),
    strip.text = element_text(size = 20),
    strip.background = element_blank(),
    panel.border = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    legend.position = "none")
```

```
package_estimates <- eff_brm.mn.package[[1]] %>%
  clean_names() %>%
  rename(
    course_pack = cats,
    probability_estimate = estimate) %>%
  select(discipline, course_pack, probability_estimate, lower, upper)
```

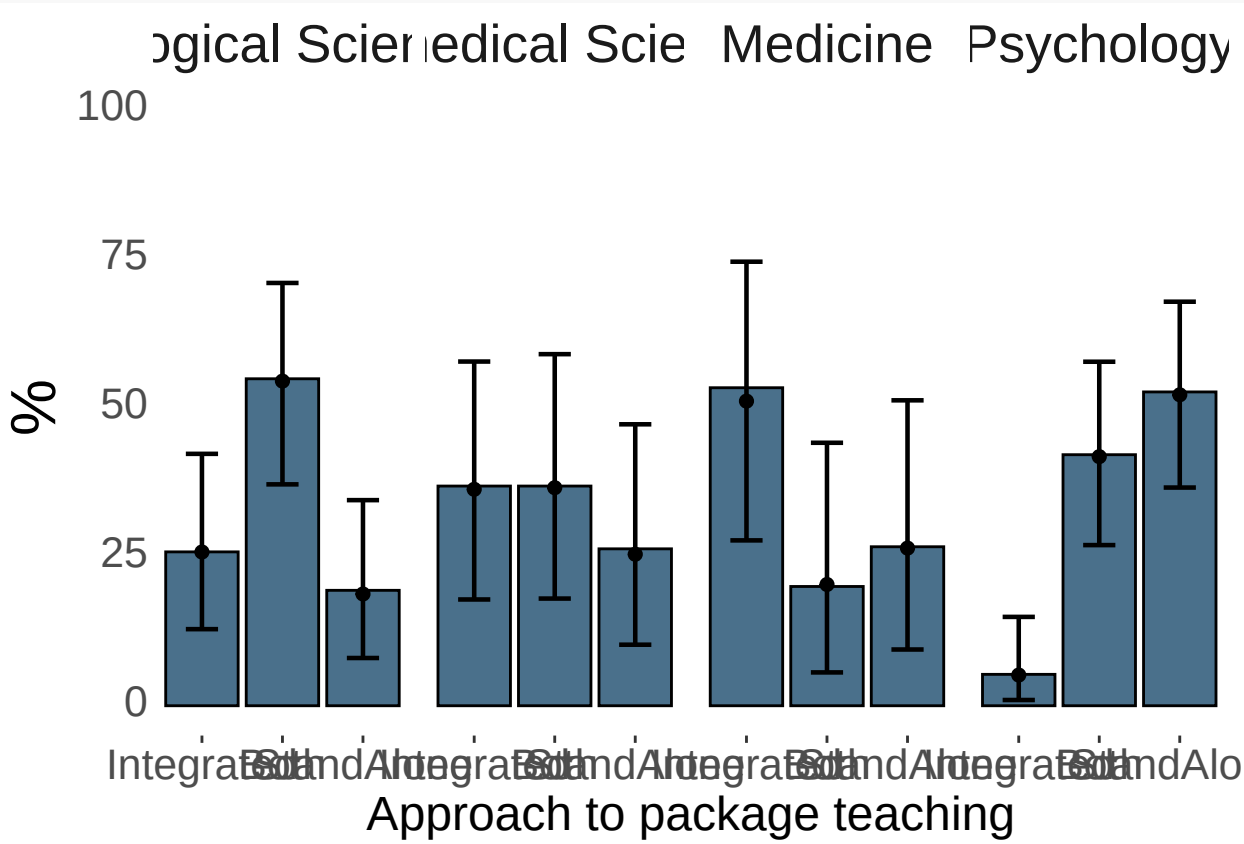
```
package_estimates
```

```
##    discipline course_pack probability_estimate      lower     upper
## 1         Bio  Integrated           0.25783060 0.12845570 0.4224629
## 2      Biomed  Integrated           0.36295933 0.17822789 0.5772315
## 3         Med  Integrated           0.51088683 0.27733554 0.7446878
## 4       Psych  Integrated           0.05147568 0.00960613 0.1490096
## 5         Bio        Both           0.54444177 0.37126957 0.7090329
## 6      Biomed        Both           0.36568004 0.17971981 0.5896562
## 7         Med        Both           0.20335370 0.05591315 0.4410128
## 8       Psych        Both           0.41777798 0.26941088 0.5769172
## 9         Bio  StandAlone           0.18740504 0.08006153 0.3448246
## 10     Biomed  StandAlone           0.25404688 0.10235398 0.4719898
## 11        Med  StandAlone           0.26437928 0.09430945 0.5122189
## 12      Psych  StandAlone           0.52150200 0.36592143 0.6775902
```

```
survey_table_pack <- table(survey_data$course_packaging, survey_data$discipline)
survey_table_pack <- prop.table(survey_table_pack, margin = 2)
survey_table_pack <- as.data.frame(survey_table_pack)
survey_ggplot_pack <- survey_table_pack %>%
  rename(
    course_pack = Var1,
    discipline = Var2) %>%
  mutate(percentage = 100*Freq)
```

```
package_plot <- ggplot() +
  geom_col(
    data = survey_ggplot_pack,
    aes(x = course_pack, y = percentage),
    color = "black",
    fill = plot_colour) +
  geom_errorbar(
    data = package_estimates,
    aes(x = course_pack, ymin = 100*lower, ymax = 100*upper),
    width = .4,
    linewidth = .8) +
  geom_point(
    data = package_estimates,
    aes(x = course_pack, y = 100*probability_estimate),
    size = 2) +
  scale_x_discrete(drop = FALSE) +
  scale_y_continuous(
    "%",
    limits = c(0, 100),
    labels = c(0, 25, 50, 75, 100)) +
  facet_wrap(
    ~ factor(
      discipline,
      labels = c("Biological Sciences", "Biomedical Sciences", "Medicine", "Psychology")),
    nrow = 1) +
  xlab("Approach to package teaching") +
  plot_theme
```

```
package_plot
```




```
ggsave(
  plot = package_plot,
  here("figures", "package_plot_gelman.jpg"),
  device = "jpeg",
  width = 1280,
  height = 637,
  units = "px",
  dpi = 100)
```

Perform Sensitivity Analysis:

```
#Sensitivity analysis:  determines the extent to which the prior affects the results
#(a qualitative test, not a quantitative test)

brm.mn.package_fixed <- tidy(brm.mn.package, effects = "fixed", conf.level = 0.95, fix.intercept = FALSE
  mutate(
    ci_width = abs(conf.low - conf.high)
  )
```

```
## Warning in tidy.brmsfit(brm.mn.package, effects = "fixed", conf.level = 0.95, :
## some parameter names contain underscores: term naming may be unreliable!
```

```
brm.mn.package_fixed
```

```
## # A tibble: 8 x 8
##   effect component term           estimate std.error conf.low conf.high ci_width
##   <chr>  <chr>     <chr>             <dbl>     <dbl>    <dbl>     <dbl>    <dbl>
```

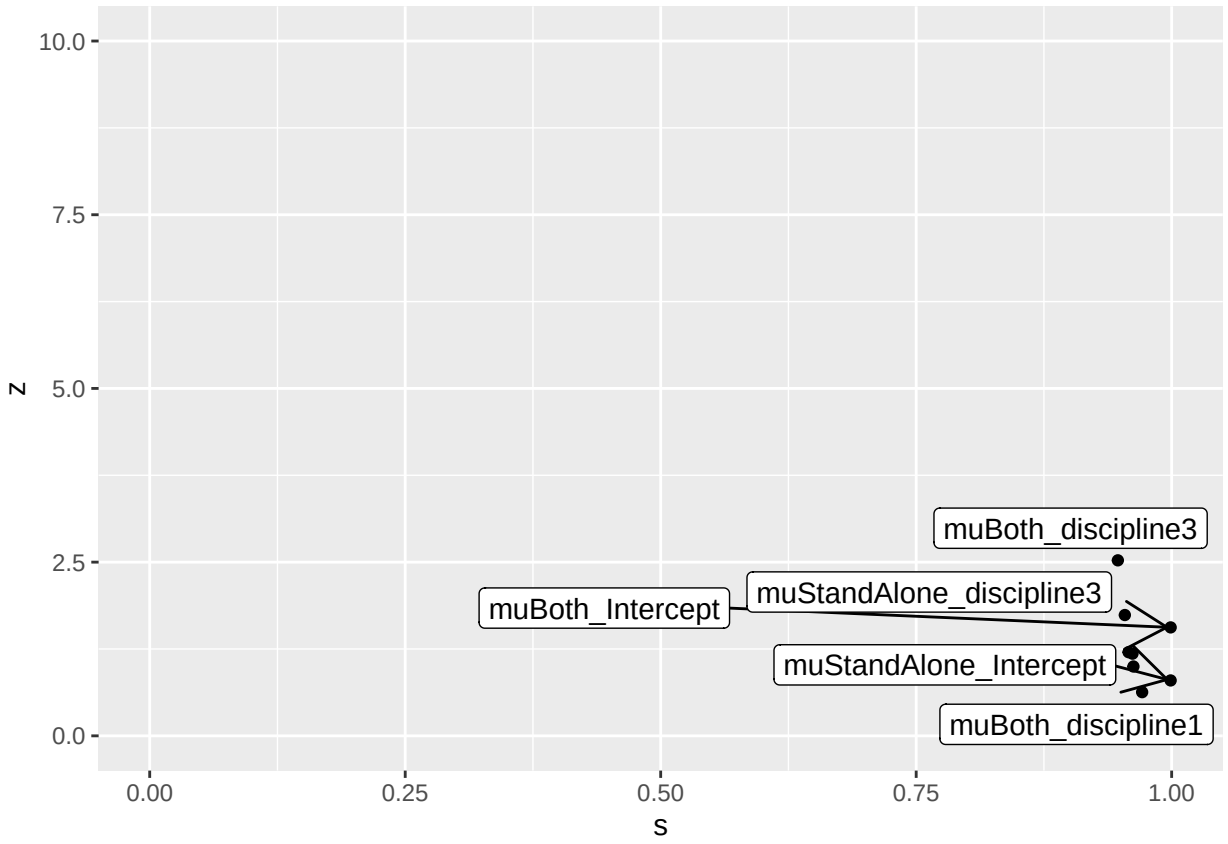
```
## 1 fixed  cond      muBoth_Interc~    0.490     0.314   -0.104    1.13       1.23
## 2 fixed  cond      muStandAlone_~    0.254     0.319   -0.352    0.911      1.26
## 3 fixed  cond      muBoth_discip~    0.267     0.425   -0.557    1.12       1.68
## 4 fixed  cond      muBoth_discip~   -0.483     0.484   -1.44     0.447      1.89
## 5 fixed  cond      muBoth_discip~   -1.45      0.574   -2.64    -0.382      2.26
## 6 fixed  cond      muStandAlone_~   -0.579     0.489   -1.56     0.374      1.93
## 7 fixed  cond      muStandAlone_~   -0.618     0.513   -1.65     0.377      2.03
## 8 fixed  cond      muStandAlone_~   -0.930     0.535   -2.02     0.0907     2.11
```

```
#We cannot see all the term names, so we'll reduce to essential components:
brm.mn.package_fixed.reduced <- brm.mn.package_fixed %>%
  select(effect, component, term, estimate
  )
brm.mn.package_fixed.reduced
```

```
## # A tibble: 8 x 4
##   effect component term                     estimate
##   <chr>  <chr>     <chr>                       <dbl>
## 1 fixed  cond      muBoth_Intercept            0.490
## 2 fixed  cond      muStandAlone_Intercept      0.254
## 3 fixed  cond      muBoth_discipline1          0.267
## 4 fixed  cond      muBoth_discipline2         -0.483
## 5 fixed  cond      muBoth_discipline3         -1.45
## 6 fixed  cond      muStandAlone_discipline1   -0.579
## 7 fixed  cond      muStandAlone_discipline2   -0.618
## 8 fixed  cond      muStandAlone_discipline3   -0.930
```

```
#We have 8 fixed effects; the first 2 are intercepts, and the last 6 are coefficients (b)
#Also, the 2 Intercepts are listed first in the output of brm.mn.package_fixed,
#and the Intercepts had priors with sd = 10
#The remaining 6 fixed effects are for b, which had sd = 2.5 for the prior.
#Hence, we set up the sensitivity analysis like this:
brm.mn.package_fixed %>%
  mutate(
    theta = c(0, 0, 0, 0, 0, 0, 0, 0),
    sigma_prior = c(10, 10, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5),
    z = abs((estimate - theta) / std.error), # it's called here std.error but is the standard deviation
    s = 1 - (std.error^2 / sigma_prior^2)
  ) %>%
  ggplot(aes(s, z, label = term)) +
  geom_point() +
  geom_label_repel(arrow = arrow()) +
  xlim(0, 1) + ylim(0, 10)
```

```
## Warning: ggrepel: 3 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps
```


10.0
7.5
5.0
2.5
0.0
z
0.00
0.25
0.50
0.75
1.00
s
muBoth_discipline3
muStandAlone_discipline3
muBoth_Intercept
muStandAlone_Intercept
muBoth_discipline1